

# Practice Quiz 05C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 05

*Contrapositive or contradiction; a universally quantified goal.*

Write in full sentences. Introduce the arbitrary element and name your proof strategy.

1. Prove

$$\forall x \in \mathbf{Z} \ (25 \nmid x^2 \rightarrow 5 \nmid x).$$

**Solution** Let  $x$  be an arbitrary integer. We prove the contrapositive

$$5 \mid x \longrightarrow 25 \mid x^2.$$

Suppose  $5 \mid x$ . Then  $x = 5k$  for some integer  $k$ . Therefore

$$x^2 = (5k)^2 = 25k^2.$$

Because  $k^2$  is an integer,  $25 \mid x^2$ . This proves the contrapositive, so  $25 \nmid x^2 \rightarrow 5 \nmid x$ . Since  $x$  was arbitrary, the universally quantified statement follows. ■